

Yash Shukla

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Research Interests

- Reinforcement Learning (RL)
- Robotics / Robot Learning
- Foundational Models (LLMs/VLMs)
- 2D / 3D Generative AI (Diffusion Models)
- Sequential Decision Making
- Computer Vision / Deep Learning (CV/DL)
- Natural Language Processing (NLP)
- Task and Motion Planning (TAMP)

Education

- 2020 – 2025 **Ph.D. in Computer Science**
GPA: 3.8/4.0 Tufts University, Medford, MA (Advisor: [Jivko Sinapov](#))
Relevant Courses: Reinforcement Learning, Probabilistic Robotics, Algorithms, Software Engineering
- 2020 – 2025 **Master of Science in Robotics Engineering**
GPA: 4.0/4.0 Worcester Polytechnic Institute (WPI), Worcester, MA
Relevant Courses: Deep Learning, Artificial Intelligence, Robot Control, Human Robot Interaction
- 2014 – 2018 **Bachelor of Engineering (Hons.) in Mechanical Engineering**
GPA: 8.4/10.0 Birla Institute of Technology and Science, Pilani, India (University acceptance rate ~ 2%)
Relevant Courses: Digital Image Processing, Robotics and Mechanisms, Mechatronics

Experience

- April '25 – **Berkshire Grey**, Senior Software Engineer with the R&D Team
Present
 - Robot learning for bimanipulation robotic arms for Berkshire Grey's Robotic Pick Cells.
- June '24 – **Autodesk Research**, Research Intern with Robotics Research Team in SF, CA
Aug '24
 - Developed a large-scale (**4M Samples**) 6-DOF industrial grasp dataset using Physics-Based simulator (**Isaac Lab**).
 - Developed novel **Diffusion Modeling** technique using informative source priors for generative 6-DOF grasp synthesis.
- Oct '24 – **Mitsubishi Electric Research Labs**, Research Intern, hosted by Siddarth Jain
Apr '24
 - Developed precise **6-DOF grasp pose** generation for industrial objects using **VLMs** for object keypoint detection.
 - Utilized **multi-modal** learning through images, tactile, and proprioception to enhance **robotic manipulation**.
- Sept '23 – **Tufts University**, Lecturer for CS 138 - Reinforcement Learning
Present
 - Designing and instructing **CS 138 Reinforcement Learning** (*First student ever at Tufts to teach RL*).
 - Leading a team of graduate teaching assistants and undergraduate class assistants.
- May '23 – **Georgia Tech Research Institute**, Research Intern
Sept '23
 - Utilized **Foundational Models (LLMs, VLMs)** to suggest task plans that improve sampling efficiency of RL agents.
 - Developed novel **Nuerosymbolic AI** approaches that combine formal methods with reinforcement learning.

Aug '20 – **Tufts University**, Research Assistant

Present

- Formulating new ideas in **Generative AI, Reinforcement Learning, Imitation Learning, Neurosymbolic AI**.
- Designing **transfer learning** techniques for policy, value function, and curriculum transfer between different fidelities of robotic agents (First authored publications at AAMAS '22, SRL (ICRA) '22, Sim2Real (RSS) '22).
- Pioneering Neurosymbolic methods by integrating TAMP with RL for **open-world novelty** handling (ICDL 22).
- Enhancing sampling efficiency and generalization capacity of RL agents by integrating **Foundation Models** with reinforcement learning agents (AAMAS '24 and ICAPS '24).
- Generative AI techniques (**VAE, GANs, Diffusion Models**) for Sim2Real robot learning (IROS '23).
- Current work focuses on: LLMs for Open-World Novelty Accommodation for Robotic Manipulation

May '19 – **MathWorks**, Computer Vision Team

Aug '19

- Developed an **End-to-end differentiable** CV algorithm to improve accuracy of camera calibration parameters for Fisheye Cameras.
- Achieved better checkerboard detection precision (98%) as compared to the existing technique (83%).

Publications

- RLDM 2025 Formal and Natural Language Guided Dynamic Task Sampling for Reinforcement Learning Agents
Yash Shukla, Jivko Sinapov
Reinforcement Learning and Decision Making Conference 2025
- CoRL 2025 GraspFactory: A Large Object-Centric Grasping Dataset
Srinidhi Srinivas **Yash Shukla**, Adam Arnold, Sachin Chitta
Making Sense of Data in Robotics workshop at CoRL 2025 [\[Link\]](#)
- IROS 2024 A Framework for Neurosymbolic Goal-Conditioned Continual Learning for Open World Environments
Pierrick Lorang, Shivam Goel, **Yash Shukla**, Patrik Zips, Matthias Scheutz
International Conference on Intelligent Robots and Systems [\[Link\]](#)
- IROS 2024 Autonomous Robotic Assembly: From Part Singulation to Precise Assembly
Kei Ota, Devesh Jha, Siddarth Jain, William Yezounis, Radu Corcodel, **Yash Shukla**, Antonia Bronars, Diego Romeres
International Conference on Intelligent Robots and Systems [\[Link\]](#)
- RLC 2024 Agent-Centric Human Demonstrations Train World Models
James Staley, Shivam Goel, **Yash Shukla**, Elaine Short
Reinforcement Learning Conference [\[Link\]](#)
- ICAPS 2024 Logical Specifications-guided Dynamic Task Sampling for Reinforcement Learning Agents
Yash Shukla, Tanushree Burman, Abhishek Kulkarni, Robert Wright, Alvaro Velasquez, Jivko Sinapov
International Conference on Automated Planning and Scheduling [\[Link\]](#)
- AAMAS 2024 LgTS: Dynamic Task Sampling using LLM-generated sub-goals for Reinforcement Learning Agents
Yash Shukla, Wenchang Gao, Vasanth Sarathy, Alvaro Velasquez, Robert Wright, Jivko Sinapov
International Conference on Autonomous Agents and Multi-Agent Systems [\[Link\]](#)
- IROS 2023 A Framework for Few-Shot Policy Transfer through Observation Mapping and Behavior Cloning
Yash Shukla, Bharat Kesari, Shivam Goel, Robert Wright, and Jivko Sinapov
International Conference on Intelligent Robots and Systems [\[Link\]](#)
- ICAPS 2023 Automaton-Guided Curriculum Generation for Reinforcement Learning Agents
Yash Shukla, Abhishek Kulkarni, Robert Wright, Alvaro Velasquez, and Jivko Sinapov
International Conference on Automated Planning and Scheduling [\[Link\]](#)
- AAMAS 2022 ACuTE: Automatic Curriculum Transfer from Simple to Complex Environments
Yash Shukla, Christopher Thierauf, Ramtin Hosseini, Gyan Tatiya, Jivko Sinapov
International Conference on Autonomous Agents and Multiagent Systems [\[Link\]](#)
- ICDL 2022 RAPid-Learn: A Framework for Learning to Recover for Handling Novelities in Open-World Environments
Shivam Goel*, **Yash Shukla***, Vasanth Sarathy, Matthias Scheutz, and Jivko Sinapov
International Conference on Development and Learning
* - Denotes equal contribution [\[Link\]](#)
- RSS 2022 A Framework for Curriculum Schema Transfer from Low-Fidelity to High-Fidelity Environments
Yash Shukla, Jivko Sinapov
Closing the Reality Gap in Sim2Real Transfer for Robotics at Robotics: Science and Systems [\[Link\]](#)

- ICRA 2022 An Object-Oriented Approach for Generating Low-Fidelity Environments for Curriculum Schema Transfer
Yash Shukla, Kaleb Loar, Robert Wright, Jivko Sinapov
Scaling Robot Learning Workshop at International Conference on Robotics and Automation [\[Link\]](#)
- IROS 2020 Haptic Knowledge Transfer Between Heterogeneous Robots using Kernel Manifold Alignment
Gyan Tatiya, **Yash Shukla**, Michael Edegware, and Jivko Sinapov
International Conference on Intelligent Robots and Systems [\[Link\]](#)

Teaching

- Instructor Reinforcement Learning at Tufts University (Spring 2025, Fall 2024, Spring 2024, Fall 2023)
Teaching Assistant Big Data at Tufts University (Fall 2020)
Peer Learning Assistant Physics 101 at Worcester Polytechnic Institute (Fall 2019, Spring 2020)

Mentoring

- Tufts Shijie Fang, Wenchang Gao, Bharat Kesari, Tanushree Burman, Anirudh Nair, Wei Wo Lin

Service

- Program Committee ICAPS 2026
Reviewer TMLR, PLAN-FM at AAAI 2026, RLC 2025, PRL at ICAPS 24, IROS 2024, RLC 2024, AAMAS 2023, ICRA 2024, RA-L 2024, Artificial Intelligence for Autonomous Driving at IJCAI-ECAI 2022
Student Volunteer ICRA 2022, RLDM 2023 (Head volunteer)
Program Committee Artificial Intelligence for Autonomous Driving at IJCAI-ECAI 2022

References

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| ○ Prof. Jivko Sinapov
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Tufts University
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Senior Research Scientist, Georgia
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